



PROLOGUE — PREPARE THE DATA

One flat file becomes a star schema.

Before any visual, the data model decides whether DAX, slicers and time intelligence will work.

TOOL

Power BI · Power Query

METHOD

Kimball star schema

OUTPUT

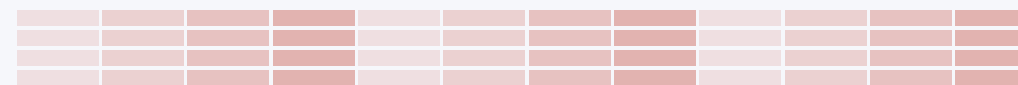
1 fact · 5 dimensions

BEFORE

flat file

57 columns · one table

Supplier name, country & tier repeat 5,200 times — once per PO line. Bloated; many-to-one impossible.



AFTER

star schema

1 lean fact + 5 dims

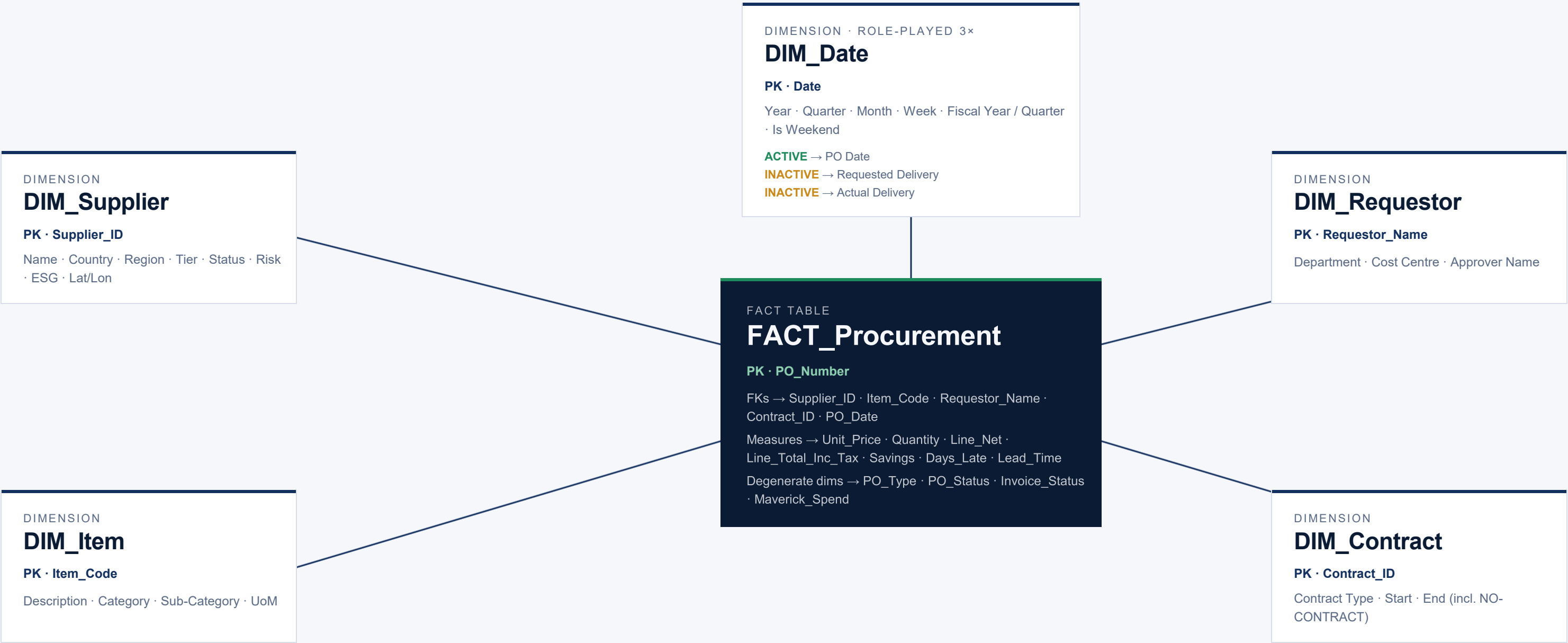
Descriptive attributes extracted to unique dimension tables. FACT holds keys + measures.



FACT · 5 DIMS

ONE FACT · FIVE DIMENSIONS · DATE ROLE-PLAYED THREE TIMES

The shape of the model decides everything downstream.



POWER QUERY · DUPLICATE → SELECT → DEDUPLICATE

Extract dimensions, then strip the fact table.

FOUR STEPS · REPEAT PER DIMENSION

01
Duplicate

02
Select cols

03
Remove others

04
Remove duplicates on PK

DIM_Supplier PK · Supplier_ID	Supplier ID · Name · Country · Region · Tier · Status · Risk · Payment Terms · Preferred · Local/Int · ESG · Lat/Lon	15 unique rows
DIM_Item PK · Item_Code	Item Code · Description · Category · Sub Category · Unit of Measure	~50 unique rows
DIM_Requestor PK · Requestor_Name	Requestor Name · Department · Cost Centre · Approver Name	15 unique rows
DIM_Contract PK · Contract_ID	Contract ID · Contract Type · Contract Start · Contract End	~3,400 unique

```
= Table.Distinct( Table.SelectColumns( Data,
{ "Supplier ID", "Supplier Name", "Country",
  "Tier", "Status", "Risk", "ESG Score", ... } ) )
```

FACT TABLE · KEEP VS REMOVE

REMOVE — NOW IN DIM_SUPPLIER	Name · Country · Region · Tier · Status · Risk · Payment Terms · ESG · Lat/Lon
REMOVE — NOW IN DIM_ITEM	Description · Category · Sub-Category · UoM
REMOVE — NOW IN DIM_REQUESTOR	Department · Cost Centre · Approver Name
REMOVE — NOW IN DIM_CONTRACT	Contract Type · Start · End
REMOVE — USE DIM_DATE	PO Year · PO Quarter · PO Month
KEEP — FOREIGN KEYS	PO Number · PO Date · Requested / Actual Delivery · Supplier_ID · Item_Code · Requestor · Contract_ID
KEEP — MEASURES	Unit Price · Quantity · Line Net · Line Total Inc Tax · Savings · Days Late · Lead Time Days
KEEP — DEGENERATE DIMS	PO Type · PO Status · Invoice Status · Match Type · Maverick · Single Source

NOTE

Never use the raw flat table as a dimension — Supplier_ID repeats thousands of times and breaks many-to-one. Always deduplicate first.

RELATIONSHIPS · GOLDEN RULES · STARTER DAX

Wire it once, correctly — the rest is measures.

RELATIONSHIPS · ALL MANY-TO-ONE

FROM (DIMENSION)	TO (FACT)	ACTIVE
DIM_Date	FACT[PO_Date]	ACTIVE
DIM_Date	FACT[Requested_Delivery]	INACTIVE
DIM_Date	FACT[Actual_Delivery]	INACTIVE
DIM_Supplier	FACT[Supplier_ID]	ACTIVE
DIM_Item	FACT[Item_Code]	ACTIVE
DIM_Requestor	FACT[Requestor_Name]	ACTIVE
DIM_Contract	FACT[Contract_ID]	ACTIVE

DIMENSION GOLDEN RULES

01 · UNIQUE PRIMARY KEY Every dimension table MUST have one unique row per primary key.	02 · VERIFY AFTER TABLE.DISTINCT Row count of DIM must equal distinct IDs in FACT.
03 · NO REDUNDANT COLUMNS IN FACT Once descriptive data lives in a dim, drop it from the fact.	04 · DEGENERATE DIMS STAY IN FACT Low-cardinality flags work directly as slicers.

STARTER DAX MEASURES

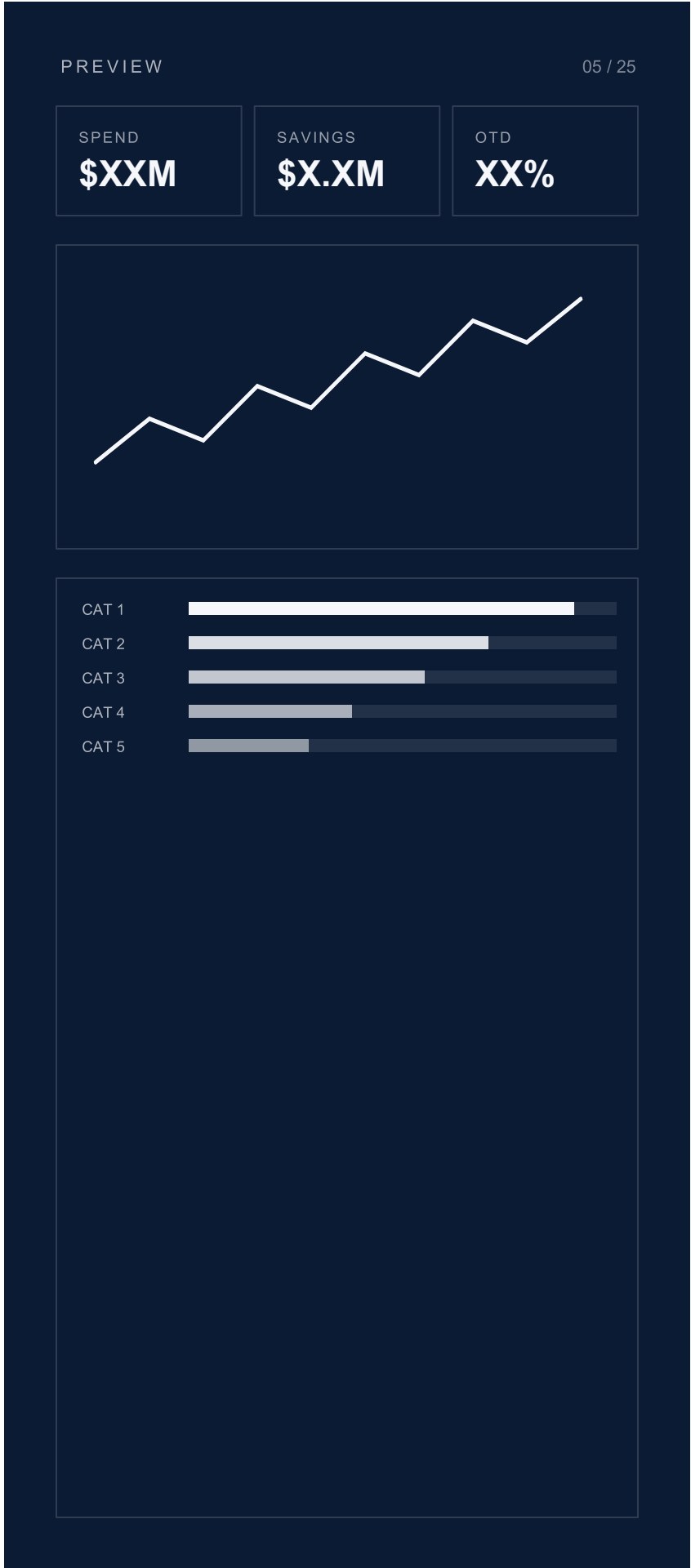
TOTAL SPEND SUM(FACT_Procurement[Line_Total_Inc_Tax])
OTIF % DIVIDE(COUNTROWS(FILTER(FACT, [On_Time_Delivery] = "Yes")), COUNTROWS(FACT), 0)
MAVERICK % DIVIDE(CALCULATE([Total Spend], FACT[Maverick_Spend] = "Yes"), [Total Spend], 0)
SPEND BY ACTUAL DELIVERY CALCULATE([Total Spend], USERRELATIONSHIP(FACT[Actual_Delivery], DIM_Date[Date]))

TIP

Mark DIM_Date as Date Table (Table Tools → Mark as Date Table). Required for TOTALYTD, SAMEPERIODLASTYEAR and all time intelligence.

Executive Procurement Overview.

A build guide for challenge participants: what to show, why it matters, and how to make it readable in ten seconds.



SIX QUESTIONS · SIX KPIS

Every KPI answers a question.

CORE BUSINESS QUESTIONS

Q 1	How much are we spending?
Q 2	Are we within budget?
Q 3	What savings have been achieved?
Q 4	Is spend controlled through preferred suppliers?
Q 5	Are suppliers delivering on time?
Q 6	Which categories drive the most spend?

TOP KPI ROW · RECOMMENDED ORDER

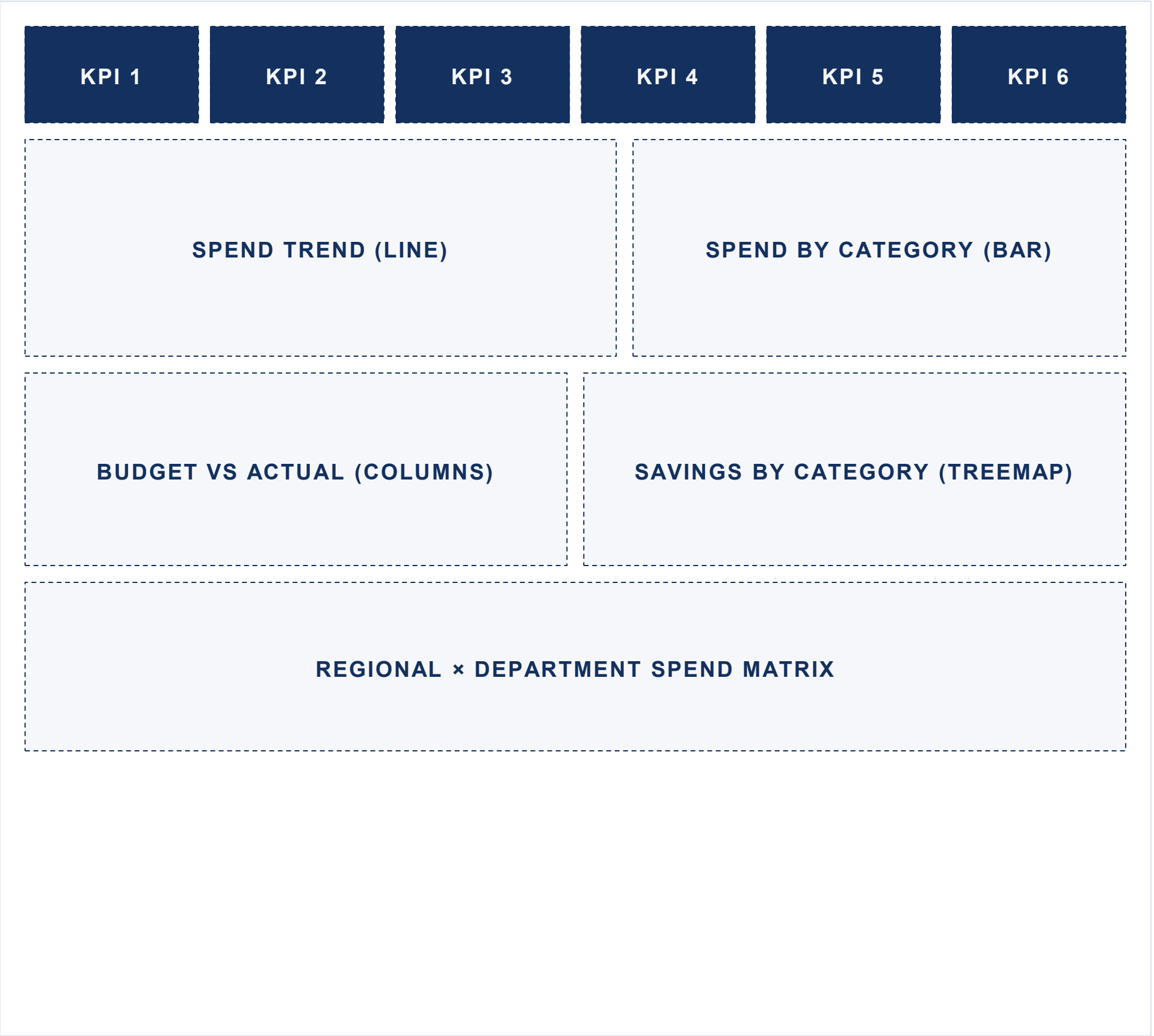
K 1	Managed Spend	\$XXX.XM
K 2	Savings Delivered	\$XX.XM
K 3	Budget Variance	-X.X%
K 4	Active Suppliers	X,XXX
K 5	Preferred Spend %	XX%
K 6	On-Time Delivery %	XX%

RECOMMENDED LAYOUT

Stack it in four zones, top to bottom.

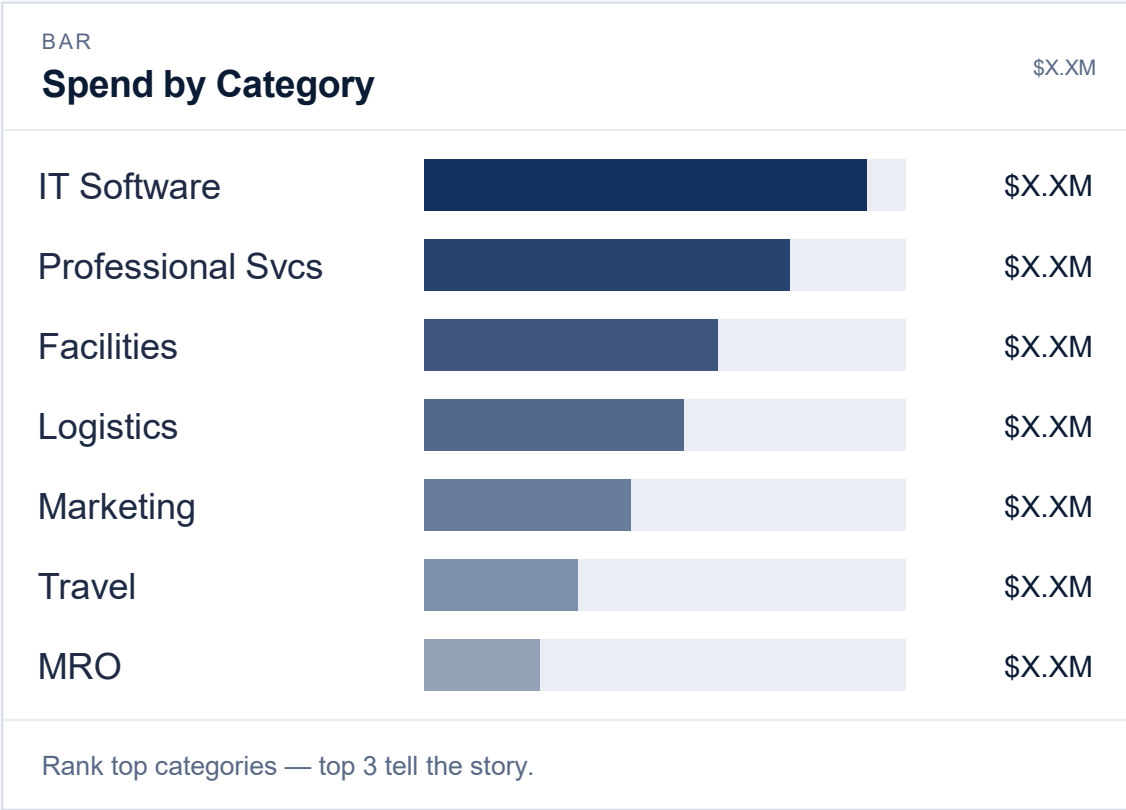
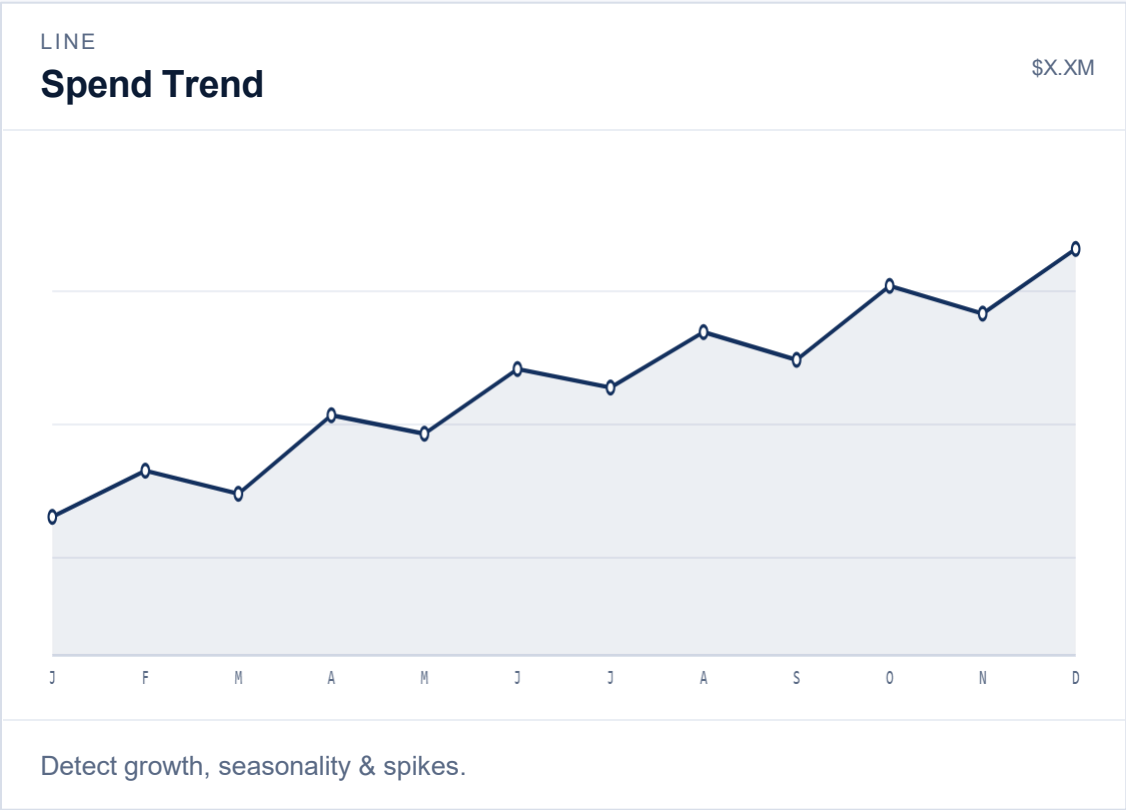
Four horizontal bands — KPI row, trend + category, budget + savings, regional matrix.

- 1 KPI row across the top
- 2 Trend + category breakdown
- 3 Budget vs actual + savings
- 4 Region × department matrix



FIVE VISUALS · ONE PER QUESTION

Five visuals carry the page.



RULE OF THUMB

One chart per question. If a chart has no question, cut it.

Data labels + ranking colors

EXECUTIVE INSIGHT CALLOUTS · DYNAMIC TEXT

Say it in one sentence — or don't say it at all.

Short, dynamic sentences sit under each visual. Tone matches the status color — one sentence per callout, no hedging.

• POSITIVE

Spend remains below budget with healthy savings delivery.

• OBSERVATION

IT Software is the highest spend category.

• GOVERNANCE

Preferred supplier usage is improving procurement governance.

• WATCH

Delivery performance requires attention in selected regions.

DESIGN DIRECTION · WINNING TIPS

Keep the canvas quiet and the palette confident.

PALETTE

Canvas	Navy	Good	Warn	Risk
#FFFFFF	#13305E	#1F8A5B	#C78A1B	#C0392B

TYPE

Inter Tight · Display

Inter Tight · Body 400

JetBrains Mono · Meta / Numbers

Clean white canvas · Navy headers · Green = positive · Amber = warning · Red = risk · Minimal clutter, executive readability.

WINNING TIPS

- 01

Prioritize clarity over quantity
- 02

One chart per business question
- 03

Add short insights, not just charts
- 04

Balance finance × ops × governance
- 05

Ten-second readability test

NORTH STAR

Make this page understandable in 10 seconds.

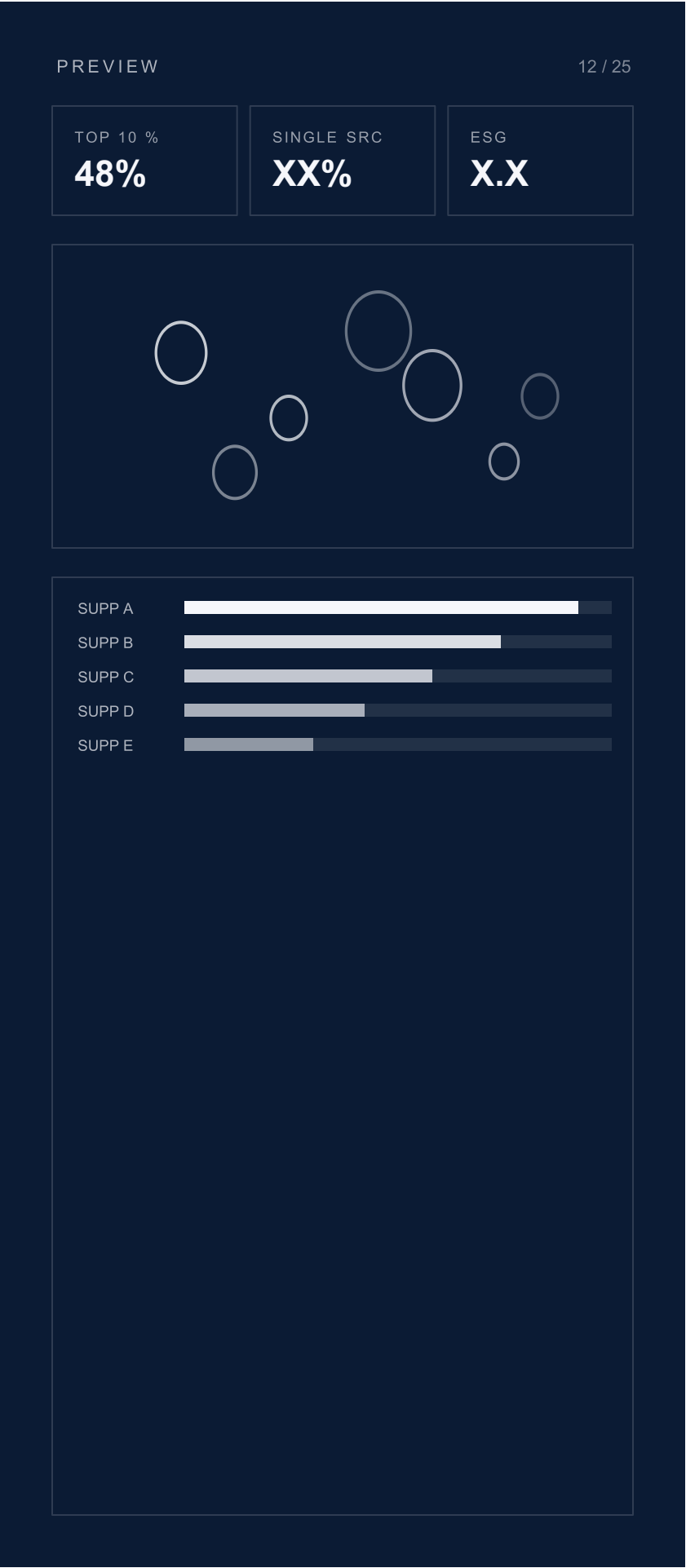
SUMMARY

Build it in this order — you're done with Page 1.

QUESTIONS	KPI ROW	VISUALS	INSIGHTS
6	6	5	4
business questions answered up front	metrics · finance × ops × governance	one per question · nothing extra	dynamic callouts under the visuals

Supplier Intelligence & Risk.

Are we buying from the right suppliers, in the right way, with acceptable risk?



SEVEN QUESTIONS · SIX KPIS

Every supplier KPI answers a governance question.

CORE BUSINESS QUESTIONS

Q 1	Which suppliers receive the highest spend?
Q 2	How concentrated is supplier spend?
Q 3	Are preferred suppliers being used effectively?
Q 4	How much spend depends on single-source suppliers?
Q 5	Which suppliers carry the highest risk?
Q 6	How do suppliers compare by ESG, tier & status?
Q 7	Are strategic suppliers performing better?

TOP KPI ROW · RECOMMENDED ORDER

K 1	Supplier Spend Concentration %	48%
K 2	Preferred Spend %	XX%
K 3	Single Source Spend %	XX%
K 4	High Risk Supplier Spend	\$XX.XM
K 5	Avg ESG Score	X.X / 5
K 6	Active Strategic Suppliers	XXX

RECOMMENDED LAYOUT

Mirror Page 1 — four zones, supplier lens.

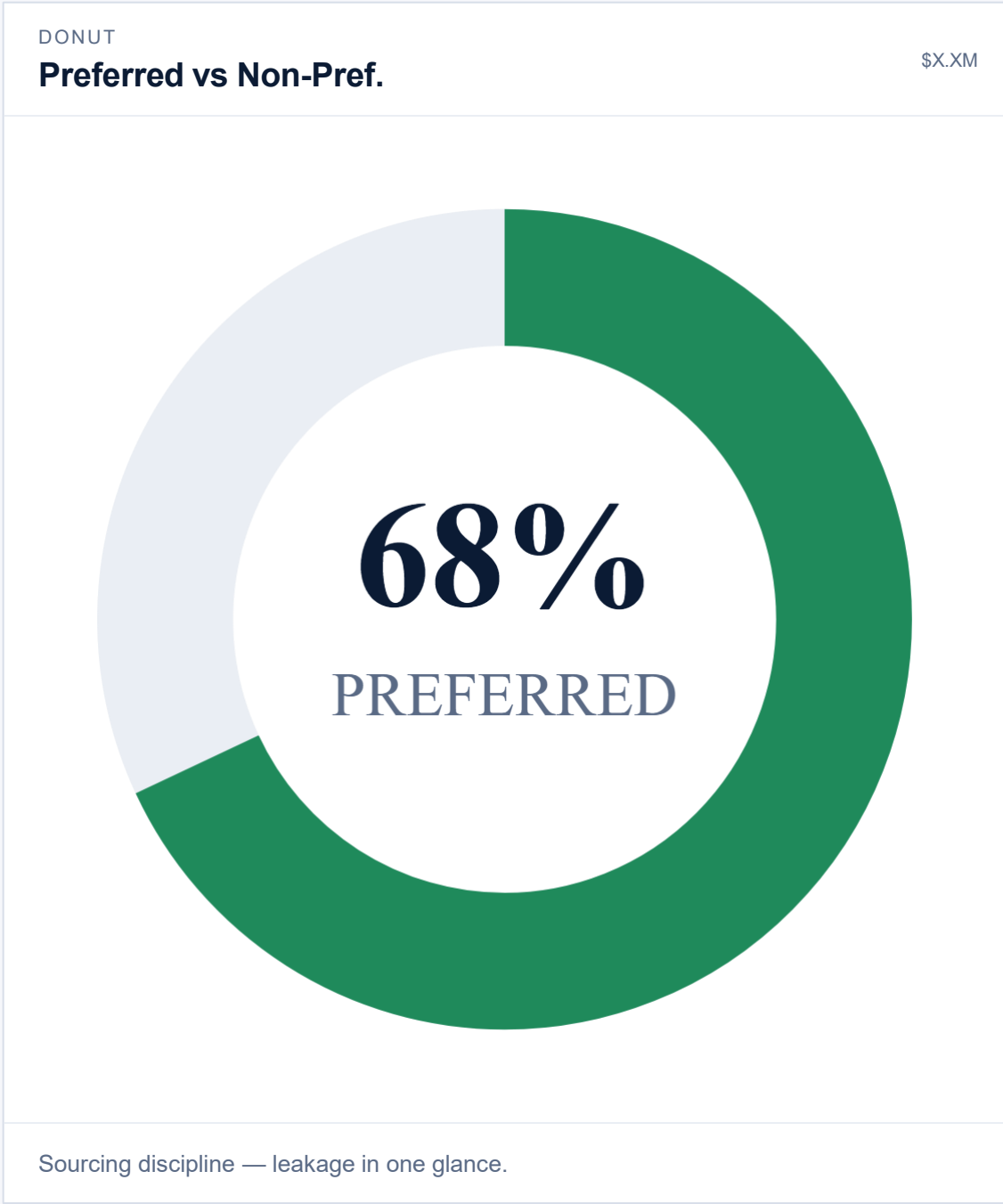
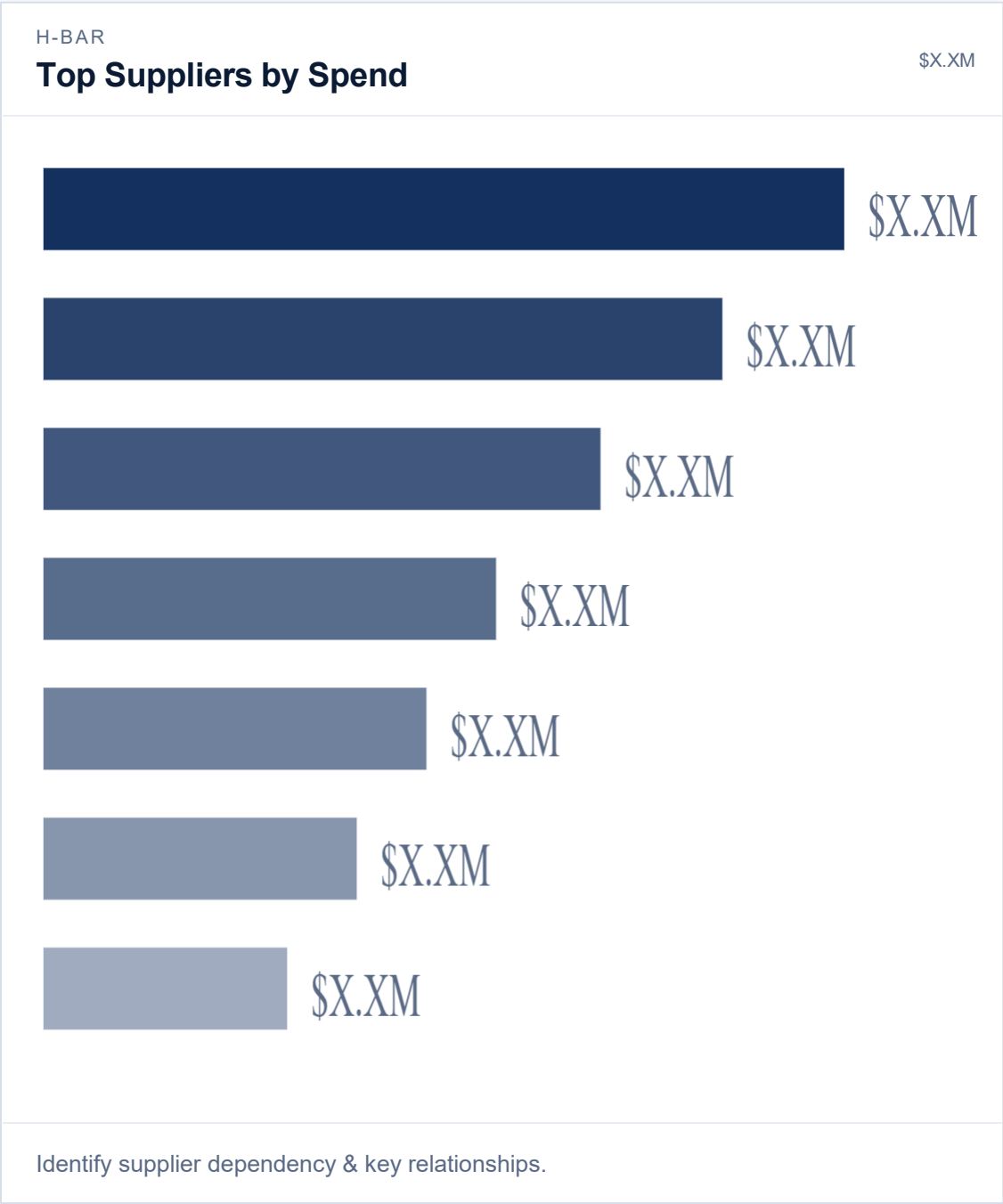
Consistency with Page 1 — KPI row, supplier ranking + risk/ESG, preferred + single source, governance matrix.

- 1 KPI row across the top
- 2 Top suppliers + risk/ESG scatter
- 3 Preferred vs non-preferred + single source
- 4 Supplier × region × status matrix



FIVE VISUALS · SUPPLIER LENS

Rank, risk, discipline, dependency, governance.



EXECUTIVE INSIGHT CALLOUTS · DYNAMIC TEXT

Surface the supplier risk story in five lines.

Pair each visual with a single sentence — tone matches the status color, no hedging.

• CONCENTRATION

Top 10 suppliers represent 48% of total managed spend.

• LEAKAGE

Non-preferred suppliers still account for significant spend leakage.

• RISK

High-risk suppliers concentrated in Logistics and IT Hardware.

• ESG

ESG scores strongest among strategic Tier 1 suppliers.

• DEPENDENCY

Single-source dependency remains elevated in selected categories.

MESSAGE

Strong procurement is not only about spend — it is about choosing the right suppliers with resilience, governance, and long-term value.

DESIGN DIRECTION · WINNING TIPS

Same visual style as Page 1 — new semantics.

PALETTE · REPURPOSED

Canvas	Navy	ESG / Pref	Dependency	High Risk
#FFFFFF	#13305E	#1F8A5B	#C78A1B	#C0392B

ICONOGRAPHY

Icons for risk · ESG · supplier tier

Keep set small — reuse across both pages.

White canvas · Navy headers · Green = preferred / strong ESG · Amber = dependency · Red = high risk.

WINNING TIPS

- 01

Limit to Top 10 / Top 15 suppliers
- 02

Use scatter to show risk intelligently
- 03

Show preferred vs non-preferred clearly
- 04

Single-source risk is gold — surface it
- 05

Balance spend with ESG and quality
- 06

Make concentration risk immediate

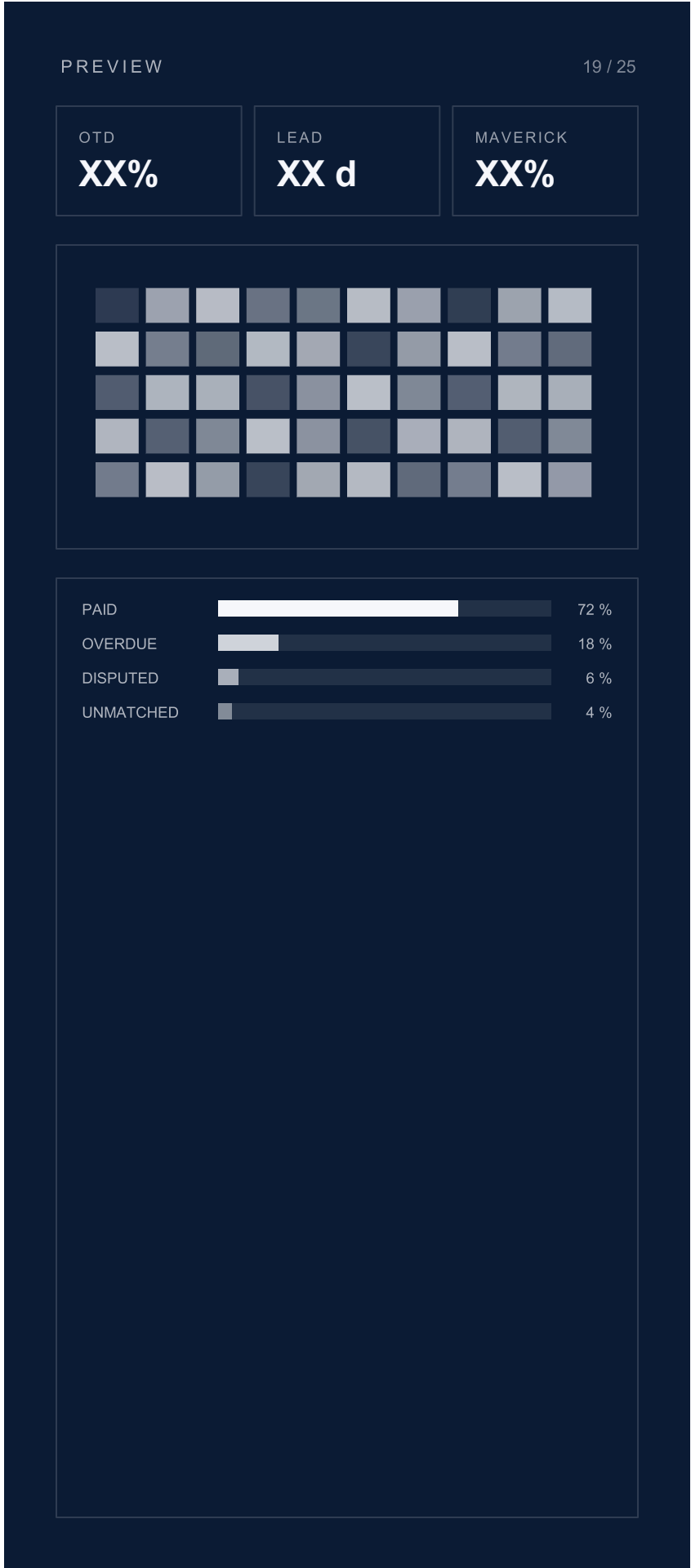
SUMMARY

Right suppliers · right way · acceptable risk.

QUESTIONS	KPI ROW	VISUALS	INSIGHTS
7	6	5	5
supplier questions answered up front	concentration × risk × ESG	rank · scatter · donut · bar · matrix	dynamic callouts, one per finding

Control Tower & Efficiency.

How efficiently and in control is the procurement operation?



SEVEN QUESTIONS · SIX KPIS

Every operational KPI answers a control question.

CORE BUSINESS QUESTIONS

Q 1	What is the overall on-time delivery rate?
Q 2	Which suppliers, categories or departments delay most?
Q 3	How do lead times vary across suppliers / categories?
Q 4	Is maverick spend increasing — and where?
Q 5	What share of invoices are overdue, disputed or unmatched?
Q 6	Where are the main operational warning signs?
Q 7	Which areas need urgent procurement intervention?

TOP KPI ROW · RECOMMENDED ORDER

K 1	On-Time Delivery %	XX%
K 2	Avg Lead Time	XX d
K 3	Maverick Spend %	XX%
K 4	Overdue Invoices %	XX%
K 5	Disputed Invoices %	X.X%
K 6	3-Way Match %	XX%

RECOMMENDED LAYOUT

Same four zones — operational lens.

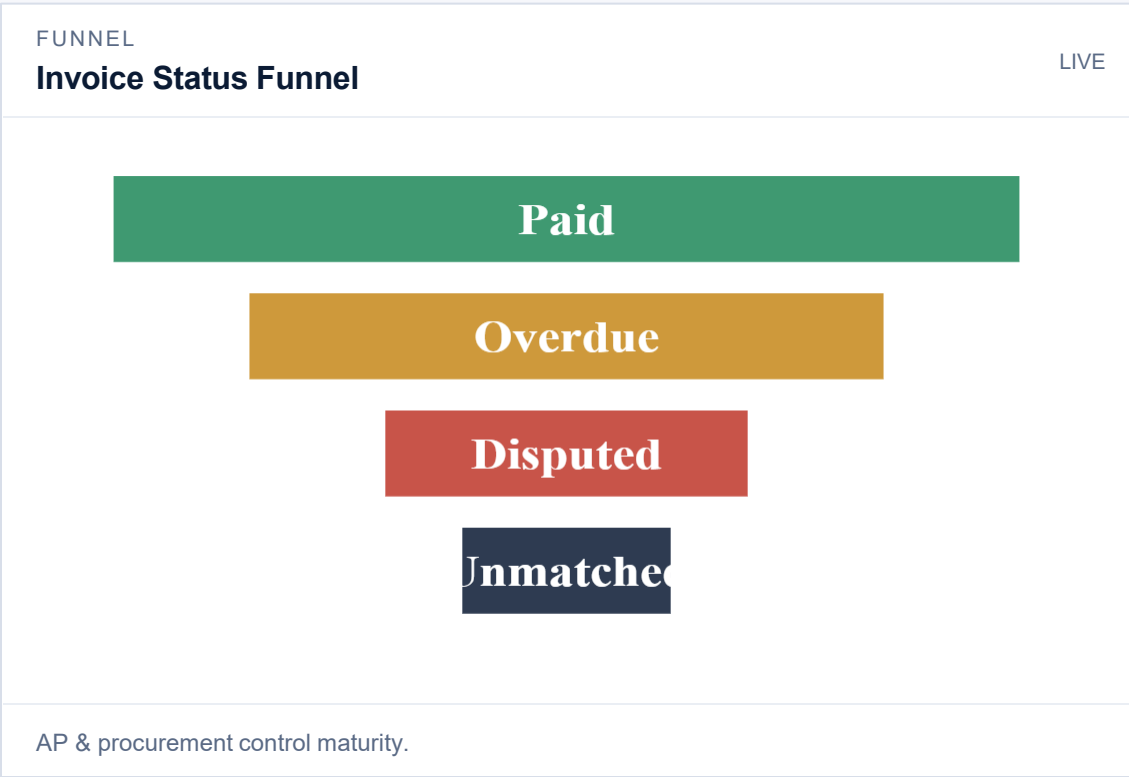
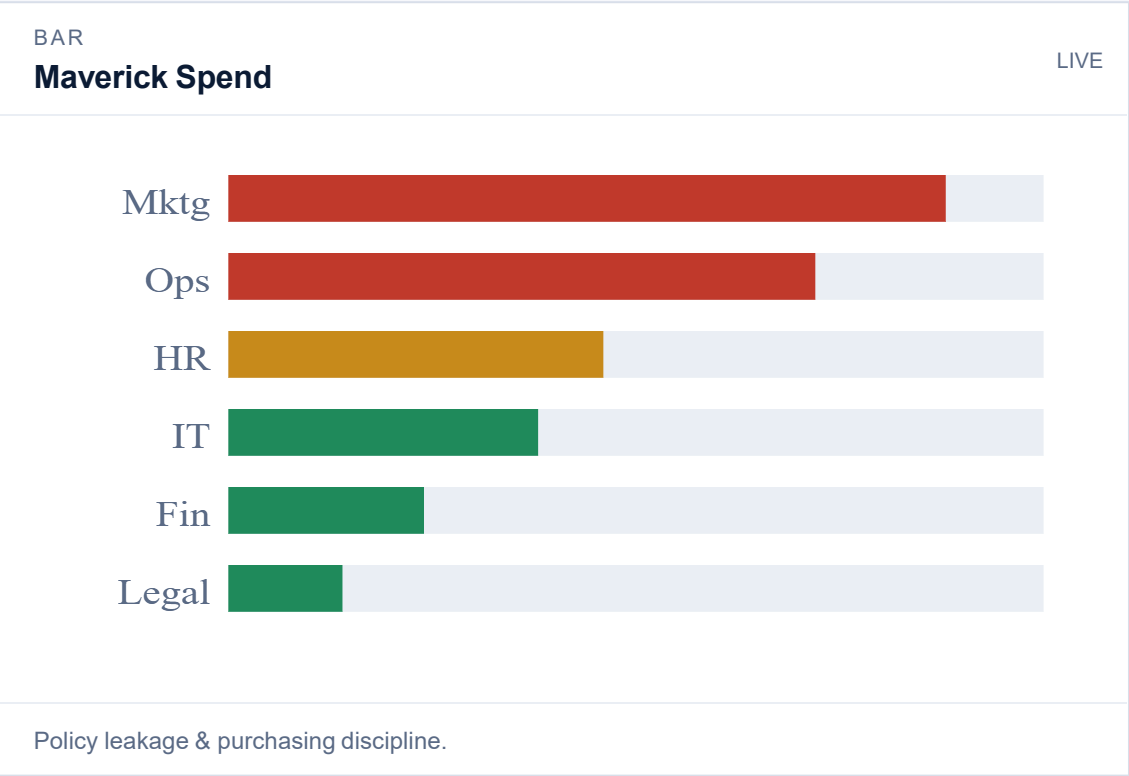
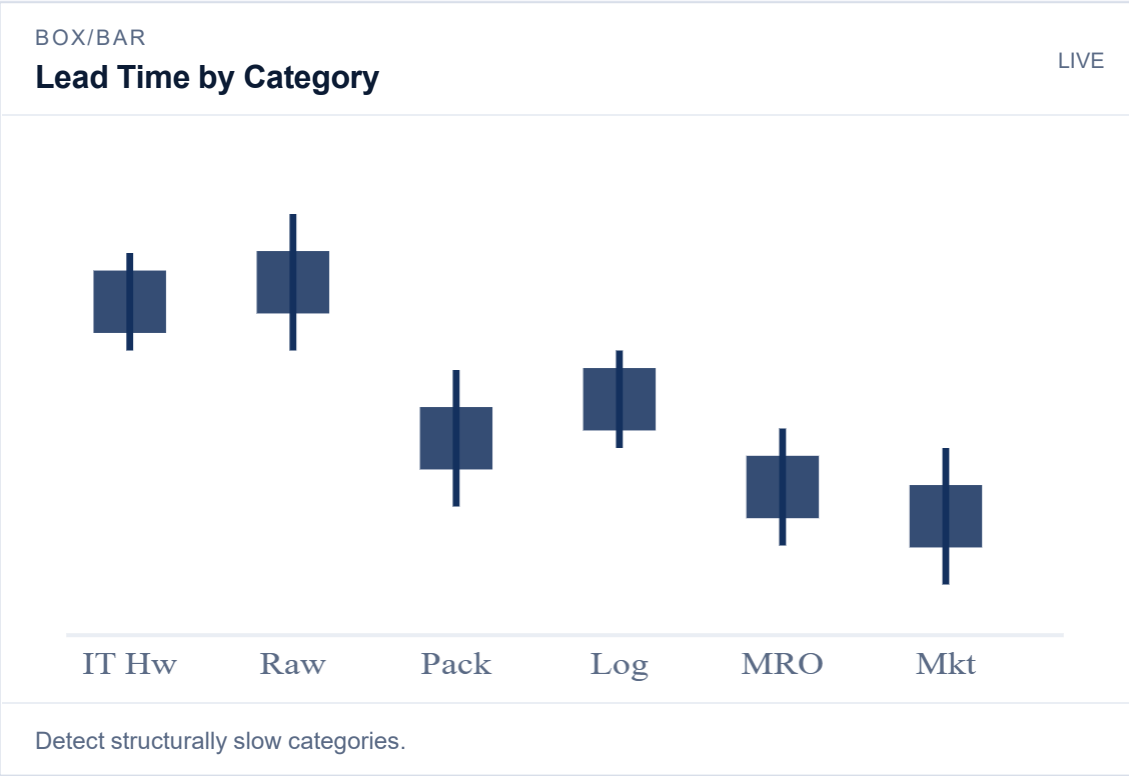
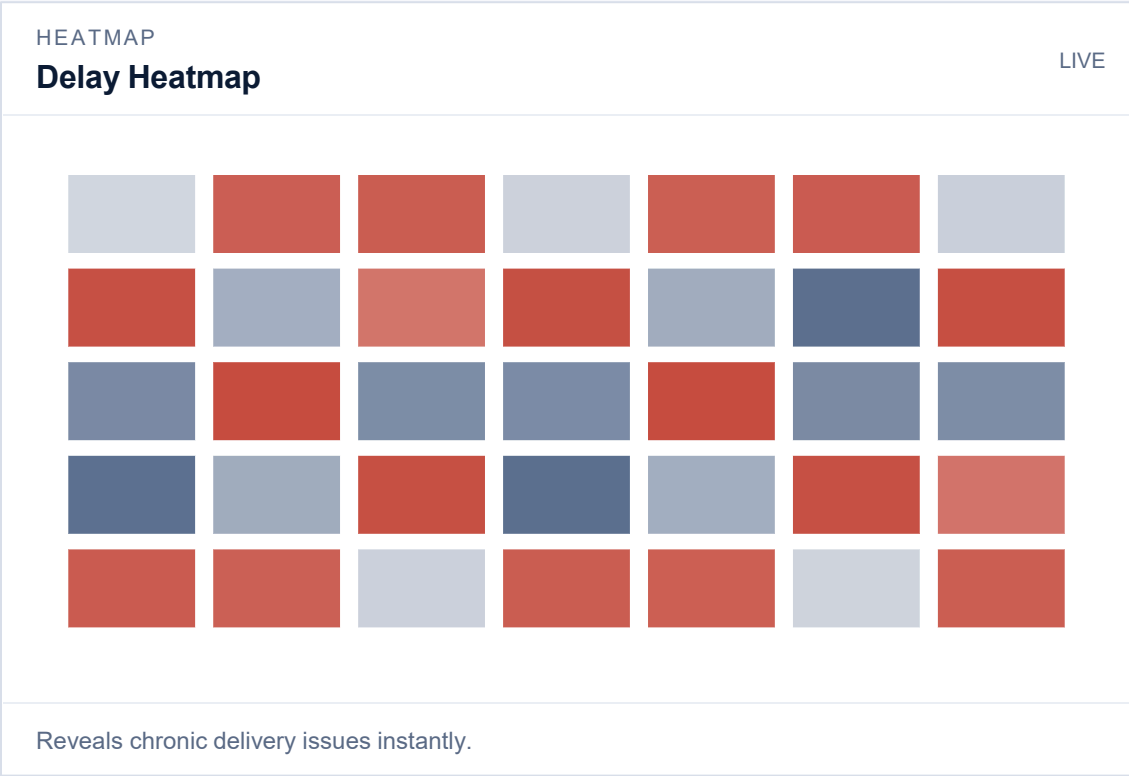
Mirror Pages 1 & 2 — KPI row, delay + lead time, maverick + invoices, warning dashboard.

- 1 KPI row across the top
- 2 Delay heatmap + lead time
- 3 Maverick spend + invoice funnel
- 4 Warning dashboard / risk summary



FIVE VISUALS · CONTROL LENS

Delay · lead time · leakage · flow · warnings.



RULE OF THUMB

Heatmaps for delays. Trends over totals. Leadership-actionable callouts.

Finance × Operations × Compliance

EXECUTIVE INSIGHT CALLOUTS · DYNAMIC TEXT

Surface operational warnings leadership can act on.

One sentence per finding — tone matches status color — action-oriented language only.

• DELIVERY

On-time delivery below target due to selected suppliers.

• LEAKAGE

Maverick spend highest within Marketing and Operations.

• DISPUTES

Invoice disputes concentrated in Professional Services.

• LEAD TIME

Longest lead times found in IT Hardware and Raw Materials.

• CONTROL

3-way match compliance improving — still below best practice.

MESSAGE

Procurement success is not only what you buy or from whom — it is how reliably, compliantly and efficiently the process operates every day.

DESIGN DIRECTION · WINNING TIPS

Consistent with Pages 1 & 2 — urgency visible.

PALETTE · OPERATIONAL SEMANTICS

Canvas #FFFFFF	Navy #13305E	On Target #1F8A5B	Warning #C78A1B	Urgent #C0392B
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ICONOGRAPHY

Alerts · clocks · invoice symbols

Reuse the Page 2 icon set — extend, don't reinvent.

White canvas · Navy headers · Green = on target · Amber = warning · Red = urgent operational issue.

WINNING TIPS

- 01

Heatmaps for delays — powerful & efficient
- 02

Show trends, not only totals
- 03

Maverick spend is an executive KPI
- 04

Combine finance + operations + compliance
- 05

Add clear warning callouts
- 06

Finish with action-oriented storytelling

SUMMARY · END OF BLUEPRINT

Three pages · one executive procurement story.

QUESTIONS	KPI ROW	VISUALS	INSIGHTS
7	6	5	5
operational questions answered up front	delivery × compliance × flow	heatmap · box · bar · funnel · tiles	action-oriented dynamic callouts